

Ordinal Profile-LRT Finite-Sample Calibration

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Purpose

This note narrows the profile-LRT small-sample problem to all-ordinal limited-information SEM. The target is not a universal Bartlett correction. The target is a decision rule for profile-LRT confidence intervals for polychoric omega under DWLS first, with WLS as a later comparison, where thresholds, polychorics, NACOV, and fitted weights are estimated from categorical margins.

The companion note [1] records the general distinction: when the limiting reference is a weighted chi-square spectrum, a scalar Bartlett factor is only a first-moment correction. Here the focal profile restriction has one degree of freedom. After robust or misspecification scaling, the leading reference again has the form χ_1^2 , so a finite-sample scalar is at least meaningful. The question is whether it is useful, stable, and diagnostically separable from ordinal point-estimate bias.

1 Object

Let s collect ordinal sample thresholds and polychoric correlations, and let $F(\theta; s, W)$ be the limited-information DWLS or WLS criterion. For a scalar functional $g(\theta)$, here polychoric/latent-response omega, define

$$T(g_0) = N\{F(\tilde{\theta}(g_0); s, W) - F(\hat{\theta}; s, W)\}, \quad \tilde{\theta}(g_0) = \arg \min_{\theta: g(\theta)=g_0} F(\theta; s, W).$$

The current magmaan profile helpers also compute one-degree-of-freedom reference scalings at the constrained endpoint. Write the reference-scaled

statistic as

$$Z(g_0) = \begin{cases} T(g_0), & \text{ordinary,} \\ T(g_0)/\lambda_{\text{rob}}, & \text{robust scaled,} \\ T(g_0)/\lambda_{\text{mis}}, & \text{misspecification scaled.} \end{cases}$$

The asymptotic reference for $Z(g_0)$ is χ_1^2 . The finite-sample question is whether

$$Z(g_0) \leq c \chi_{1,1-\alpha}^2$$

with a stable c improves coverage in the small ordinal cells.

2 Bias Decomposition

The ordinal problem has four separable failure modes.

1. *LR inflation.* Even after the robust endpoint scaling, $\mathbb{E}Z$ may exceed one at small N . This is the only component a Bartlett-type threshold factor directly targets.
2. *Reference-scale error.* The endpoint λ is itself estimated from thresholds, polychorics, NACOV, and sometimes fitted weights. Sparse cells can move the robust or misspecification reference before any Bartlett factor is considered.
3. *Functional point bias.* The unrestricted omega estimate may be biased because the polychoric stage and the fitted CFA surface are biased at small N . A threshold correction can cover the population value at g_0 , but it cannot make a biased point estimate a good center for width summaries.
4. *Boundary and feasibility mass.* For bounded reliability coefficients, some samples may put the profile surface near a boundary or make g_0 hard to attain. In the maximal-reliability experiments this mass dominated the mean inflation and defeated analytic Lawley and per-sample bootstrap factors.

3 Candidate Corrections

Oracle cell constant. For a design cell, estimate $c_{\text{oracle}} = \mathbb{E}Z(g_{\text{pop}})$ by Monte Carlo and report coverage under $Z(g_{\text{pop}}) \leq c_{\text{oracle}} \chi_{1,1-\alpha}^2$. This is not a method, but it identifies whether the statistic is repairable by a scalar.

Ordinal null-bootstrap constant. For a design class, simulate ordinal data from the constrained fitted model imposing $g = g_0$, refit the unrestricted and constrained models, and average the reference-scaled Z . The value is used as a cell-level constant, not as a per-sample random factor. The earlier maximal-reliability result rules out per-sample use: the bootstrap factor was anti-correlated with the samples that needed correction most.

Shrinkage constant. If the null-bootstrap constant is noisy but directionally useful, shrink it to a grid prior:

$$c_{\text{shrink}} = w c_{\text{boot}} + (1 - w) c_{\text{grid}}.$$

The weight w should be chosen by coverage and anti-correlation diagnostics, not by mean recovery alone.

Weight and NACOV regularization. If reference-scale error dominates, the correction target is not c but the estimated ordinal Γ and W . DLS, EBA/pEBA-style spectrum shrinkage, or regularized NACOV can be tested as inputs to the profile reference. These are finite-sample reference corrections, not Bartlett factors.

Bias-aware reporting. If point-functional bias dominates, report the profile interval as calibrated for test inversion but keep a separate bias diagnostic for ω . Do not hide functional bias inside the Bartlett factor.

4 Experiment 53

Experiment 53 is the first taxonomy run. It uses all-ordinal one-factor probit data, DWLS fits, and the existing `frontier_profile_lrt_ordinal_polychoric_omega` surface. It records ordinary, robust-scaled, and misspecification-scaled profile tests at the population ω , plus the fit-free robust delta interval for comparison. The initial grid deliberately defers the WLS robust-scaled arm to spend the simulation budget on DWLS replications.

The decision rule is:

1. If uncorrected Z is near χ_1^2 and delta coverage is acceptable, there is no ordinal Bartlett problem in that cell.
2. If Z is inflated and the oracle constant fixes coverage, advance to an ordinal null-bootstrap constant.

3. If Z is not inflated but coverage is poor, prioritize omega point-bias and sparse-threshold diagnostics.
4. If robust and misspecification references diverge materially, keep the small-sample constant conditional on the selected reference regime.

The first full DWLS-only grid used 1000 replications per cell over $N \in \{50, 100, 250, 500\}$ and balanced versus threshold-extreme cuts. Ordinary DWLS profile inversion is not a viable reference here: coverage stayed around 0.60–0.69 because the unscaled statistic has mean roughly 3.5–6.4. Endpoint robust scaling and the misspecification-scaled reference both move the reference coverage close to nominal in most cells, roughly 0.95–0.97. The exception is the threshold-extreme $N = 50$ cell, where empty or near-empty categories create profile failures and unstable fitted-weight reference scales; coverage is about 0.93 rather than 0.95. The practical next correction target is therefore sparse ordinal-summary / fitted-weight stability in the small extreme-threshold cell, not a broad Bartlett correction for every ordinal omega interval.

The misspecified extension adds a locally dependent latent-response DGP, with residual correlations between items 1–2 and 5–6, while fitting the same one-factor DWLS model. The target is the large-sample pseudo-true model-implied profile omega from the working one-factor fit, not the generating model’s reliability. With 1000 replications per cell and pseudo-targets from 100000 generated cases, ordinary DWLS again undercovers badly (about 0.59–0.69). Endpoint robust scaling covers about 0.94–0.97, and the misspecification-scaled reference covers about 0.94–0.97. The weakest misspecified cell remains threshold-extreme $N = 50$, with many profile failures and unstable fitted-weight scales. Thus the misspecification reference is behaving as intended for the working-model target, but it does not remove the sparse ordinal-summary problem.

The scalar misspecification reference should be read as scaled, not as a substantive mixture distinction. For a one-degree-of-freedom profile test the weighted- χ^2 law has a single eigenvalue λ , so comparing T to $\lambda\chi_1^2$ is equivalent to comparing T/λ to χ_1^2 . The experiment keeps any older “mixture” spelling only as a legacy alias for this scalar reference.

The stress extension adds two harder working-model-misspecification cells: stronger residual local dependence and a two-factor population, both still fit by the same one-factor DWLS model. The stress default uses $N \in \{25, 35, 50, 75\}$ and threshold-extreme / threshold-sparse cuts. The sparse-threshold cells mostly diagnose ordinal summary construction rather than reference-law calibration: empty categories dominate the failures. Among

cells with at least 200 successful profile fits in the 300-replication pointwise run, the useful hard cases are threshold-extreme $N = 50$. Robust-scaled coverage was 0.932 under strong local dependence and 0.926 under the two-factor DGP; the oracle-constant diagnostic moved these to 0.964 and 0.961, respectively. The misspecification-scaled arm showed the same scalar-calibration signal in the strong local-dependence cell (0.935 to 0.960) and a smaller one in the two-factor cell (0.948 to 0.967). These are the first cells where a finite-sample scalar constant is plausibly worth testing.

The first CI-inversion pass used the existing profile-CI root finder with 200 replications per cell and the same pseudo-targets. Robust-scaled inversion is the stable interval lane: among 3168 attempted robust-scaled CIs, only 38 failed, all target-test/CI inclusion decisions agreed, and cell coverage ranged from about 0.918 to 0.975. Misspecification-scaled inversion is not yet a stable CI method: among 3154 attempted CIs under the legacy singleton-mixture spelling, 1201 failed, mostly because the endpoint robust profile quadratic was not positive. Successful misspecification-scaled CIs cover near nominal, but that is a selected subset. Thus the current connected-CI evidence supports robust-scaled inversion; the misspecification-scaled path is still a pointwise reference diagnostic until endpoint reference construction is hardened.

A targeted stress CI pass on threshold-extreme $N = 50$, with 100 replications per hard DGP, preserves the same split. Robust-scaled CIs remain connected: 7 of 92 strong-local attempts and 18 of 94 two-factor attempts failed, successful coverage was 0.929 and 0.934, and the pointwise-test / CI inclusion decisions never disagreed. Misspecification-scaled endpoint inversion is still not a usable interval lane in these sparse hard cells: 178 of 179 attempted CIs failed, primarily because the endpoint robust profile sandwich quadratic was not positive. The immediate CI target is therefore robust-scaled inversion plus a candidate scalar finite-sample constant in the $N = 50$ threshold-extreme stress cells.

Working Verdict

The plausible contribution is not “a Bartlett correction for categorical SEM”. The plausible contribution is a finite-sample calibration taxonomy for ordinal profile-LRT reliability intervals. A scalar constant remains on the table only when the reference-scaled one-degree-of-freedom statistic is the source of miscoverage. Otherwise the correction belongs to the ordinal summary stage or to the reliability target itself.

References

- [1] Magmaan note. Profile likelihood-ratio intervals, the Bartlett correction, and the eigenvalue-spectrum reference law. `docs/research/notes/profile_lrt_bartlett_spectral.tex`.
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